Q.1 Prove that $\sqrt{3}$ is irrational.

Soln. Suppose

$$\sqrt{3} = \frac{p}{q}$$

where p, q are relatively prime integers.

On squaring

$$3 = \frac{p^2}{q^2}$$

So

$$p^2 = 3q^2 \quad \text{---} (1)$$

Hence $3 | p^2$ which implies $3 | p$

Let $p = 3k$

$$3k^2 = 3q^2 \quad \text{---from} (1)$$

$$\therefore q^2 = 3k^2$$

Thus ... $3 | q$

$\therefore$ P and q are divisible by 3.

Hence $\sqrt{3}$ is irrational.

Q.2 Prove that $3+\sqrt{2}$ is irrational.

Soln→

Suppose $3+\sqrt{2}$ is rational

Since, 3 is rational,

$$\sqrt{2} = (3+\sqrt{2}) - 3$$

The difference between two rational numbers must be rational

But $\sqrt{2}$ is irrational.

The statements are contradiction

$\therefore$ $3+\sqrt{2}$ is irrational.

Q.3 Solve the following using mathematical induction

(a) Sum of first n natural numbers prove:

$$1+2+3+\ldots+n = \frac{n(n+1)}{2}$$

Soln→ step - 1

Base case $n=1$

$L.H.S = 1$

$$R.H.S = \frac{1(1+1)}{2} = \frac{2}{2} = 1$$

Hence true.

Step -2 Assume true for $n = k$

$$1 + 2 + \dots + k = \frac{k(k+1)}{2}$$

step 3 - prove for $k+1$

$$1 + 2 + \dots + k + (k+1)$$

using the assumption -

$$= \frac{k(k+1)}{2} + (k+1)$$

$$= \frac{k(k+1) + 2(k+1)}{2}$$

$$= \frac{(k+1)(k+2)}{2}$$

This is required formula for $k+1$

$$\therefore 1 + 2 + \dots + n = \frac{n(n+1)}{2}$$

(b) Prove:

$$1^2 + 2^2 + \cdots n^2 = \frac{n(n+1)(2n+1)}{6}$$

Soln =

Assume for $n = k$

$$1^2 + 2^2 + \cdots + k^2 = \frac{k(k+1)(2k+1)}{6}$$

for $k+1$

$$\frac{k(k+1)(2k+1)}{6} + (k+1)^2$$

Taking $(k+1)$ common

$$(k+1)\left[\frac{k(2k+1)}{6} + (k+1)\right]$$

$$\frac{(k+1)(2k^2 + 7k + 6)}{6}$$

$$\frac{(k+1)(k+2)(2k+1)}{6}$$

Hence proved

$$\boxed{1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}}$$

(c) prove that :

$$1^3 + 2^3 + \dots + n^3 = \left[\frac{n(n+1)}{2}\right]^2$$

Assume -

$$1^3 + 2^3 + \dots + k^3 = \left[\frac{k(k+1)}{2}\right]^2$$

for $k+1$:

$$\left[\frac{k(k+1)}{2}\right]^2 + (k+1)^3$$

$$= (k+1)^2 \left[\frac{k^2}{4} + k + 1\right]$$

$$= (k+1)^2 \frac{(k+2)^2}{4}$$

$$\left[\frac{(k+1)(k+2)}{2}\right]^2$$

Hence proved.

Q.4 Prove the following

(a) $\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}$

$\therefore \frac{a}{b}$

Multiply numerator & denominator by d

$\frac{a}{b} = \frac{ad}{bd}$ ①

$\therefore \frac{c}{d} = \frac{bc}{bd}$ ② (Multiply by b)

Add ① & ②

$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}$

$= \frac{ad + bc}{bd}$

Hence proved

$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}$

(b) Same as above

$\frac{a}{b} - \frac{c}{d} = \frac{ad - bc}{bd}$

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Q. Show that $\frac{3}{5}$ and $\frac{21}{35}$ are equivalent rational numbers.

Soln →

Using the condition $ad = bc$

Here $a = 3$, $b = 5$, $c = 21$ and $d = 35$

as per $ad = bc$

$\therefore 3 \times 35 = 5 \times 21$

$\therefore 105 = 105$

$\therefore$ LHS = R.H.S

Hence - $\frac{3}{5}$ and $\frac{21}{35}$

$\therefore$ the two rational numbers are equivalent.

imp

Q. Solve $\frac{7}{12} = \frac{14}{24}$ are equivalent

Q. Solve $\frac{5}{8} = \frac{15}{20}$ are equivalent

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(cross-multiplication)

Q. Consider $\frac{a}{b}, \frac{c}{d} \in Q$ $\quad b \neq 0, d \neq 0$

prove that $\frac{a}{b} = \frac{c}{d} \iff ad = bc$

→ Proof -

we prove both

part 1 - if $\frac{a}{b} = \frac{c}{d}$ then $ad = bc$

since $b \neq 0$ & $d \neq 0$ we can multiply both sides by bd

$$bd\left(\frac{a}{b}\right) = bd\left(\frac{c}{d}\right)$$

$\therefore$ $\quad ad = bc$

$\therefore$ $\quad \frac{a}{b} = \frac{c}{d} \iff ad = bc$

part 2 - If $ad = bc$ then $\frac{a}{b} = \frac{c}{d}$

$\therefore$ Divide both sides by bd

$$\frac{ad}{bd} = \frac{bcd}{bd}$$

$\therefore$ $\frac{a}{b} = \frac{c}{d}$ $\quad \therefore$ $\frac{a}{b} = \frac{c}{d} \iff ad = bc$

Q. Prove that : if $a|b$ and $a|c$ prove that $a|b+c$ and $a|(b-c)$

Soln →

Since $a|b$ let $b = ax$ for $x \in Z$ ①

Since $a|c$ let $c = ay$ for $y \in Z$ ②

then Add ① & ②

$$b + c = ax + ay$$

$$= a(x+y) \quad \therefore$$

Similarly

$$b - c = ax - by$$

$$= a(x-y)$$

$x+y \in Z$ and $x-y \in Z$

$\therefore a|b+c$ and

$a|b-c$

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a. Solve the following - find the prime factorization

1) 24 -

Divide by the smallest prime numbers

$24 = 2 \times 12$

$= 2 \times 2 \times 6$

$= 2 \times 2 \times 2 \times 3$

$\therefore 24 = 2^3 \times 3$

This is unique prime factorization of 24

2) 36 -

$36 = 2 \times 18$

$= 2 \times 2 \times 9$

$= 2 \times 2 \times 3 \times 3$

$\therefore 36 = 2^2 \times 3^2$

3) 180 -

2×90

$2 \times 2 \times 45$

$2 \times 2 \times 3 \times 15$

$2 \times 2 \times 3 \times 3 \times 5$

$\therefore 180 = 2^2 \times 3^2 \times 5$

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Q. Find the GCD of 252 and 105

→ $252 = 105(2) + 42$

$105 = 42(2) + 21$

$42 = 21(2) + 0$

The last non-zero remainder is 21

$\therefore \boxed{\gcd(252, 105) = 21}$

Q. Find the GCD of 867 and 255

→ $867 = 255(3) + 102$

$255 = 102(2) + 51$

$102 = 51(2) + 0$

The last non-zero remainder is 51

$\therefore \boxed{\gcd(867, 255) = 51}$

Q. Find GCD of 391 and 299

Extended Euclidian [illegible]

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$\boxed{gcd(a,b) = ax + by}$

Q. Find GCD and Bezout coefficient of 48 and 18 (Extended Euclidean Algorithm)

Sol? → First we use Euclid's Algorithm

$48 = 18(2) + 12$

$18 = 12(1) + 6$

$12 = 6(2) + 0$

$\therefore gcd(48, 18) = 6$

Now we have to express 6 interms of 48 and 18

From we get $18 = 12 + 6$

From first equation, $6 = 18 - 12$ ... (1)

$12 = 48 - 18(2)$

Substitute this 2 into the previous equation

$6 = 18 - (48 - 18(2))$

$\therefore 6 = 18 - 48 + 2(18)$

$6 = -48 + 3(18)$

Hence $6 = (-1)\,48 + 3(18)$

Thus $x = -1$ & $y = 3$

Verification

$48(-1) + 18(3)$

$= -48 + 54$

$= 6$

Q. Find the HCF of 24 and 36 / L.C.M

Soln → Prime factorization of 24

$24 = 2 \times 12$

$= 2 \times 2 \times 6$

$= 2 \times 2 \times 2 \times 3$

$\therefore 24 = 2^3 \times 3$

prime factorization of 36

$36 = 2 \times 18$

$= 2 \times 2 \times 9$

$= 2 \times 2 \times 3 \times 3$

$\therefore 36 = 2^2 \times 3^2$

For HCF take the smallest powers of common prime factors

$HCF = 2^2 \times 3$

$= 4 \times 3$

$\boxed{HCF = 12}$

LCM of 24 and 36 we have

$24 = 2^3 \times 3$

and

$36 = 2^2 \times 3^2$

For LCM take the highest power of all factors

$$LCM = 2^3 \times 3^2$$
$$= 8 \times 9$$
$$= 72$$

LCM = 72

Q. Find the HCF and LCM of 60 and 90
(ans - 30, 180)

Q. Find the HCF and LCM of 40 and 56
(ans - 8, 28)

Q.2 Find GCD and Bezout Identity for 252 and 105

Soln- Euclid's Algorithm

$252 = 105(2) + 42$

$105 = 42(2) + 21$

$42 = 21(2) + 0$

$\therefore \gcd(252, 105) = 21$

Back substitution

From $105 = 42(2) + 21$

we get $21 = 105 - 2(42)$

From first equation

$42 = 252 - 2(105)$

substitute

$21 = 105 - 2[252 - 2(105)]$

$21 = 105 - 2(252) + 4(105)$

$21 = 5(105) - 2(252)$

$\therefore 21 = 5(105) + (-2)252$

so Bezout coefficients are

$x = (-2)$ & $y = 5$

Verification $252(-2) + 105(5)$

$-504 + 525$

$= 21$

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Q. Find the GCD and Bezout coefficients of 391 and 299

Soln- Using Euclid's Algorithm

$$391 = 299(1) + 92$$
$$299 = 92(3) + 23$$
$$92 = 23(4) + 0$$
$$\gcd(391, 299) = 23$$

Back substitution

$$299 = 92(3) + 23 \quad (2)$$

we get $$23 = 299 - 3(92) \quad (1)$$

From the first equation

$$92 = 391 - (1)\,299$$

Substitute in (1)

$$23 = 299 - 3[391 - (1)\,299]$$

$$23 = 299 - (3)\,391 + (3)\,299$$
$$23 = (-3)\,391 + (4)\,299$$

$$x = -3, \quad y = 4$$

verify

$$867 = (-3)\,391 + (4)\,299$$
$$= -1173 + 1196$$
$$= 23$$

Q. Find the GCD and Bezout coefficients of 867 and 255

(imp) Q. Prove that [Euclid's Algorithm]
if $a = bq + r$
then $\gcd(a,b) = \gcd(b,r)$

→ Soln - For any two positive integers a and b where $a > b$, there exist unique integers q and r such that

$$a = bq + r \qquad 0 \leq r \leq b \quad - (1)$$

where
- a = dividend
- b = divisor
- q = quotient
- r = remainder

we repeatedly apply the division algorithm until the remainder becomes 0
The last non-zero remainder is the GCD
we need to prove,

$$\gcd(a,b) = \gcd(b,r)$$

Let $d = \gcd(a,b)$
then $d \mid a$ and $d \mid b$

Since $r = a - bq$ ∴ From (1)
and $d \mid a$, $d \mid b$
so, we get

$$d \mid (a - bq)$$

$$\therefore d \mid r$$

∴ d is common divisor of b and r

Conversely, let

$c = \gcd(b, r)$

then $c \mid b$

and $c \mid r$

$a = bq + r$

$\therefore c \mid a$

$\therefore$ c is common divisor of a and b

$\therefore$ common divisor of (a, b) and (b, r) are same

$\therefore \gcd(a, b) = \gcd(b r$

9. Find the GCD of 48 and 18

→ steps - To find GCD

1) Divide the larger number by the smaller no.
2) write the remainder
3) Divide the previous divisor by the remainder
4) Continue until the remainder is 0
5) The last non-zero remainder is the GCD

Here we have to find

GCD of 48 and 18

$\therefore 48 = 18(2) + 12$ - Divide 48 by 18

$18 = 12(1) = 6$ - Divide 18 by 12

$12 = 6(2) + 0$ - Divide 12 by 6

The last non-zero remainder is 6

$\therefore \boxed{\gcd(48, 18) = 6}$